

Roll No.....

MID SEMESTER EXAMINATION MARCH - 2025

Name of the Program: B.Tech. (ME)

Semester: **VI**

Name of the Course: Automotive Mechatronics

Course Code: **TME 621**

Time: 90 minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO 1
(a)	Define mechatronics and describe its role in automotive applications. Provide examples of modern vehicle systems that incorporate mechatronic principles.	
OR		
(b)	Define high-level aggregates in automobiles, providing examples. Explain their role in vehicle functionality and performance.	
Q2	(10 marks)	CO1
(a)	How mechanical, electrical, and computer engineering disciplines are integrated into the development of mechatronic systems for automobiles. Highlight the similarities and differences between these fields in this context.	
OR		
(b)	How has technological progress influenced the aesthetics and functionality of both interior and exterior trims? Trace the development of automobile trims over time. How has technological progress influenced the aesthetics and functionality of both interior and exterior trims?	
Q3	(10 marks)	CO1
(a)	Explain the function of sensors and actuators in mechatronic automobile subsystems. Provide specific examples and discuss their roles in vehicle performance.	
OR		
(b)	Draw outline the various types of actuators used in automobiles, explaining their operating principles and their roles in contemporary vehicles.	CO2
Q4	(10 marks)	CO2
(a)	Explain the function of relays in a vehicle's electrical system. Discuss why relays are favored over direct wiring for managing high-power circuits.	
OR		
(b)	Define automotive instrumentation and discuss the importance of sensors and display systems in modern vehicles.	
Q5	(10 marks)	CO2
(a)	What are the challenges faced by automotive instrumentation systems and explore how technological advancements are helping to overcome these issues.	
OR		
(b)	Explain how a fuel gauge and speedometer function in an automobile. Contrast traditional analog displays with modern digital versions.	

